

STAT DRP Final Writeup

Decision Theory and the Black–Litterman Model

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Overview

This quarter, my DRP focused primarily on decision theory, with a smaller applied component on the Black–Litterman portfolio model. The reading that shaped the project most was *Decision Theory: Principles and Approaches* by Giovanni Parmigiani and Lurdes Inoue. Across Chapters 1–9, I learned that decision theory is a framework for organizing uncertainty, preferences, losses, and evidence into a coherent rule for acting. What I found especially valuable is that the subject moves from philosophical foundations to concrete statistical procedures in a very natural way. By the end of these chapters, ideas that first appear abstract, such as coherence, utility, admissibility, or Bayes rules, start to feel like different parts of one unified language for rational decision making.

What I learned from Chapters 1–9

The early chapters gave me the conceptual foundations. The introduction frames the main question of the field: how should a rational person act when outcomes depend on uncertain states of the world? From there, the chapter on coherence develops the Dutch Book perspective, showing that if one's betting odds violate the axioms of probability, an opponent can construct a sure loss. I liked this chapter because it turns probability from a passive description of belief into an active consistency requirement.

The chapters on utility and utility in action added the second major piece. Once uncertainty is represented with probabilities, we still need a way to express what outcomes matter more than others. Expected utility theory provides that link. The von Neumann–Morgenstern framework made clear why maximizing expected utility is a natural rule under certain preference axioms, while topics such as the St. Petersburg paradox, certainty equivalents, and risk aversion showed why raw monetary value is often the wrong object to optimize. This was one of the biggest conceptual takeaways for me: good decisions are not defined only by "what happens," but by how the decision maker values what happens.

Chapters 5 and 6 were especially important because they connect subjective beliefs and utility into a single decision-theoretic system. Ramsey and Savage show how probabilities and utilities can both be recovered from preferences, rather than assumed separately from the start. The sure-thing principle and the discussion of subjective probability made a strong impression on me because they formalize how rational beliefs can arise from consistent choices. The chapter on state independence also clarified something subtle: utility assessments are often only sensible relative to a simplified or 'small world' representation of the problem. In practice, the modeling choices we make determine what is being treated as uncertain, what is being held fixed, and what counts as a consequence.

Chapters 7–9 moved from foundations into statistical decision theory. Here the key ideas were loss functions, risk, Bayes rules, minimax rules, admissibility, and shrinkage. I found this transition very useful because it showed how the earlier philosophy becomes operational. A decision function takes data and outputs an action; the loss function measures quality; the risk averages loss over the sampling distribution; and different optimality concepts compete depending on what kind of robustness or prior information we want. The chapter on admissibility helped me see why 'reasonable sounding' rules are not always defensible, while the chapter on shrinkage illustrated that better decisions can come from deliberately pulling estimates toward a prior center rather than following raw data too aggressively. That lesson ended up connecting directly to the applied part of my project.

How this connects to Black–Litterman

The applied part of my DRP used the Black–Litterman (BL) model as a portfolio-allocation example that fits naturally into a decision-theoretic way of thinking. In my presentation, I compared Markowitz mean–variance optimization (MVO) with BL. The main issue with standard MVO is that it chooses portfolio weights directly from estimated returns and covariances, so the output can become extremely sensitive to small estimation errors in expected returns. In contrast, BL starts from a prior market equilibrium view and then blends in investor views with a specified level of confidence. In my project, the views were that QQQ would outperform SPY by 3% and that GLD would have an expected return of 5%.

What made this a good decision-theory application is that BL mirrors the structure of the book very closely. The prior corresponds to an initial belief about the world, the investor views represent additional information or judgment, and the confidence parameter controls how strongly new information should change the final action. That is very similar to the broader decision-theoretic theme that rational action is produced by combining beliefs, preferences, and uncertainty in a coherent way. Viewed this way, BL is not just a finance trick. It is a practical example of how prior beliefs and new evidence can be blended to produce a more stable decision rule.

The results in my presentation supported that interpretation. Relative to MVO, the BL portfolio was more moderated in its weights, while still tilting toward favored assets. In the backtest, BL had the highest annualized return and the highest Sharpe ratio, and it also had lower volatility and drawdown than MVO. I also examined sensitivity to the confidence parameter and found the expected pattern: as confidence increased, the portfolio tilted more strongly toward QQQ and GLD, while lower confidence kept the portfolio closer to the prior. This was a helpful demonstration that the model's behavior is interpretable rather than mysterious.

Main takeaway

My biggest takeaway from this DRP is that decision theory provides a unifying perspective across probability, statistics, and optimization. Before this project, I often thought of these subjects as separate: probability described uncertainty, statistics estimated parameters, and optimization chose the best solution. Decision theory shows that these are really parts of one pipeline. We begin with uncertainty, encode what we care about through utility or loss, update beliefs using data or judgment, and then choose an action according to a principled rule.

References

Parmigiani, G., & Inoue, L. Y. T. (2009). *Decision Theory: Principles and Approaches*. John Wiley & Sons.